UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 009 · Introduction to maxima and minima for a function of two variables · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S009·7	Deck 009, slide 7 (rendered .pptx)
T009 00:07:04	AV-summary transcript 009 (“Extremas of functions”), segment start
CW U1 P31	Unit 1 class-work problem 31
ESA Dec-23 1c	Past paper: Dec 2023, question 1.c

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1. Part A · Formula sheet
2. 2 · Definitions: relative maximum, minimum and extreme value
3. 3 · Conditions and the working rule
4. Part B · Symbol glossary

A Formula sheet

2 Definitions: relative maximum, minimum and extreme value

E2.1 · Sign of the increment ESA ★

$$\Delta = f(a+h, b+k) - f(a,b) \qquad \begin{cases} < 0 \text{ for all small } h,k: & \text{maximum} \\ > 0 \text{ for all small } h,k: & \text{minimum} \end{cases}$$

GIVES The definition of an extreme value.

APPLIES Used to settle the doubtful case $rt - s^2 = 0$.

NOTE Same sign in every direction.

S009·6 · T009 00:05:23

3 Conditions and the working rule

E3.1 · Stationary points (necessary condition) ESA ★★★

$$f_x = \frac{\partial f}{\partial x} = 0, \qquad f_y = \frac{\partial f}{\partial y} = 0$$

GIVES The candidates $(a, b), (c, d), \ldots$ for extreme values.

APPLIES f differentiable; solve the two equations simultaneously.

NOTE Two-variable version of $\frac{dy}{dx} = 0$.

S009·7 · S009·8 · T009 00:06:19

E3.2 · Second derivatives used in the test ESA ★★★

$$r = \frac{\partial^2 f}{\partial x^2}, \qquad s = \frac{\partial^2 f}{\partial x\, \partial y}, \qquad t = \frac{\partial^2 f}{\partial y^2}$$

GIVES The three numbers at each stationary point.

APPLIES Evaluate at each stationary point separately.

NOTE $s = f_{xy} = f_{yx}$ (Clairaut).

S009·7 · S009·8 · T009 00:07:04

E3.3 · Hessian determinant ESA ★★★

$$rt - s^2 = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = f_{xx} f_{yy} - f_{xy}^2$$

GIVES Whether the stationary point is an extremum.

APPLIES At a stationary point.

NOTE Positive: extremum (decide by r). Negative: saddle. Zero: doubtful.

S009·7 · T009 00:07:37

E3.4 · The test (working rule, steps 3–6) ESA ★★★

$$rt - s^2 > 0,\ r < 0 \ \Rightarrow\ \text{maximum}; \qquad rt - s^2 > 0,\ r > 0 \ \Rightarrow\ \text{minimum}$$
$$rt - s^2 < 0 \ \Rightarrow\ \text{saddle point}; \qquad rt - s^2 = 0 \ \Rightarrow\ \text{doubtful}$$

GIVES The nature of each stationary point.

APPLIES After E3.1–E3.3.

NOTE $r < 0$ plays the role of $\frac{d^2y}{dx^2} < 0$ in one variable. When $rt - s^2 > 0$, r and t have the same sign, so t can be used instead.

S009·8 · S009·9 · T009 00:11:32

B Symbol glossary

Symbol	Name	Meaning	Appears in
(a, b)	Stationary point	Solution of $f_x = f_y = 0$	E3.1
$h,\ k$	Small increments	Neighbouring point $(a+h, b+k)$	E2.1
Δ	Increment of f	$f(a+h, b+k) - f(a, b)$	E2.1
r	f_{xx}	Second partial in x at the point	E3.2–E3.4
s	f_{xy}	Mixed partial at the point (not arc length)	E3.2–E3.4
t	f_{yy}	Second partial in y (not time)	E3.2–E3.4
$rt - s^2$	Hessian determinant	$f_{xx} f_{yy} - f_{xy}^2$	E3.3

Remark. The deck uses $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$; some books write A, B, C or $D = f_{xx} f_{yy} - f_{xy}^2$. The t here is a second partial derivative, not time; the s is not arc length.